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Tutorial: Extents Countries Regions

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1. Introduction

This module builds an "extent" file. An extent is a geographic area (a country, a region, a department) traced from the real borders taken from OpenStreetMap. Thanks to this extent, a chosen imagery source (provider) applies only inside that area and stops exactly at the border. It is essential for tiles that straddle several countries (a "Multiprovider" tile): each country then gets its own imagery, with no overspill from one onto the other.

Step-by-Step Workflow

EXTENT

Enter the latitude and longitude; for our Multiprovider example, that is 49 5.

Latitude :	49	Longitude :	5
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Click Advanced



Select Generate an extent (country, region)



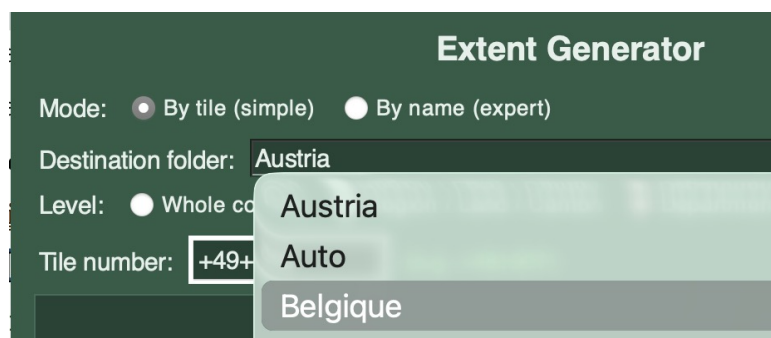
Mode line: click By tile (simple).

Mode: ☒ By tile (simple) ☐ By name (expert)

Two modes are available:

- ☒ By tile (simple): you simply enter the tile number. The module looks up on its own the countries and areas the tile crosses and lists them. This is the recommended and easiest mode.
- ☐ By name (expert): you type the exact name of the administrative area directly, as it is written in OpenStreetMap. Reserved for advanced users who already know the name they are after.

Destination folder: select Belgium for this example.



Extent Generator

Mode: ☒ By tile (simple) ☐ By name (expert)

Destination folder:

Level: ☒ Whole country ☐ Region / Land / Canton ☐ Department / Province

Tile number:

Austria

Auto

Belgique

Level line: tick Whole country.

The level sets how fine the administrative boundary used for the border will be:

- Whole country: the boundary used is that of the entire country (for example, all of Belgium).
- Region / Land / Canton: an intermediate level (a French region, a German Land, a Swiss canton, etc.).

À Department / Province: the finest subdivision (a department, a province). For everyday use, "Whole country" is enough.

Level: ☒ Whole country ☐ Region / Land / Canton ☐ Department / Province

Click List the tile's areas.

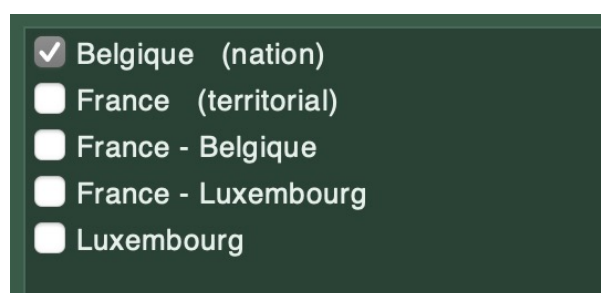


Select the tile's main country, in this case Belgium.

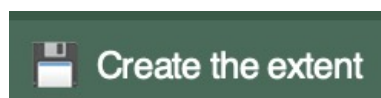
After clicking "List the tile's areas", the module shows every area that crosses the tile: the main country, but also the thin border strips and the neighbouring countries. Key point: each ticked box produces a separate extent file.

- Ticking a single area: you choose the area you want yourself, usually the tile's main country (here Belgium). The module then creates a single extent file, for that area.
- Select all: the "☒ Select all" button ticks every box at once; the module then builds, in one operation, one extent file for EACH area in the list (the main country and each of the border / neighbouring areas). The "☐ Clear all" button does the opposite.

In practice, for a Multiprovider tile, tick only the country or countries you need: each country kept gets its own extent file, to which you will then assign its imagery.



Click Create the extent



The module connects to the server automatically three times, in case the servers are busy

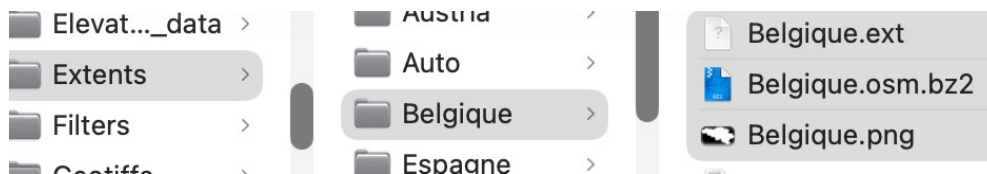
```
Creating the extent, please wait...
→ Belgium (rel[admin_level=2][name=en=Belgium])
⚡ Server busy – automatic retry (attempt 2/3)...
```

The module builds three files:

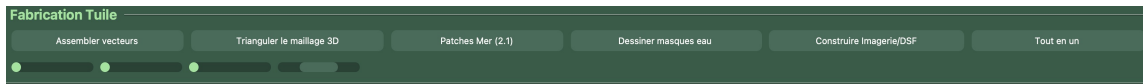
- Belgium file, .EXT
- Belgium file, .OSM.BZ2
- Belgium file, .PNG

```
Done: 1 extent(s) created.
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These files are placed automatically in the ORTHO4XP/Extents/Belgium folder.



You can now start building the tile



For experts, click Advanced settings



These three settings are reserved for experts; the default values are fine in the vast majority of cases.

- Fineness (pixel_size): precision of the area mask. The default value 1000 is a good compromise; there is no need to change it without a reason.

- Margin (buffer): preferably 0. A negative value pulls the boundary inside the border; a positive value widens it outward.
- Softening (blur): preferably 0. Creates a slight gradient at the edge. Increase it only if the imagery of the country opposite overflows onto the area.

Fineness (pixel_size)	1000
Buffer	0
Blur	0

Fineness = mask precision.
Buffer = 0 preferred (negative = pull border inward).
Blur = 0 preferred (increase only if the opposite provider overlaps).